Avelumab Monotherapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Treatment of adult patients with metastatic Merkel cell carcinoma who have received 1 or more lines of chemotherapy for metastatic disease	C44	00535a	ODMS 01/05/2019
First-line maintenance treatment of adult patients with locally advanced or metastatic urothelial carcinoma (UC) who are progression-free following platinum based chemotherapy	C65, C66, C67, C68	00535b	ODMS 01/09/2022

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

Avelumab is administered on day 1 of a 14 day cycle. Treatment should continue until disease progression or unacceptable toxicity, whichever occurs first.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	Avelumab	800mg	IV infusion	250mL NaCl 0.9% over 60 minutes	Every 14 days
Administer the solution for infusion using a sterile, non-pyrogenic, low-protein binding 0.2 micrometre in-line or add-on filter.					

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

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ELIGIBILITY:

- Indication(s) as above
- ECOG 0-1
- Adequate organ function
- **Merkel cell carcinoma**
  - Histologically proven metastatic merkel cell carcinoma
- **Urothelial carcinoma: First line**
  - Histologically confirmed, unresectable locally advanced or metastatic transitional cell carcinoma of the urothelium
  - Completed platinum based chemotherapy in the previous 4 - 10 weeks

CAUTION:

- Any medical condition that requires immunosuppressive doses of systemic corticosteroids or other immunosuppressive medication(s) (defined as >10mg prednisoLONE/daily (or steroid equivalent, excluding inhaled or topical steroids)
- Symptomatic CNS metastases
- Any active clinically significant infection requiring therapy

EXCLUSIONS:

- Hypersensitivity to avelumab or any of the excipients
- Information regarding prior therapy with an anti PD-1 or anti PD-L1 antibody is Available on the NCCP website
- History of organ transplant
- Pregnancy
- Breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Blood glucose
- Thyroid function test

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Regular tests:

- FBC, renal and liver profile prior to each cycle
- Blood glucose and thyroid function test prior to each cycle

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.
- Dose escalation or dose reduction is not recommended.
- Dosing delay or discontinuation may be required based on individual safety and tolerability

Renal and Hepatic Impairment:

Table 1: Dose modification of avelumab in renal and hepatic impairment

Renal Impairment		**Hepatic Impairment**	
CrCl mL/min	**Dose**	**Impairment level**	**Dose**
≥ 30	No dose adjustment is needed	Mild	No dose adjustment is needed
< 30	No need for dose adjustment is expected	Moderate and severe	No need for dose adjustment is expected
Haemodialysis	No need for dose adjustment is expected		
Renal and hepatic dose modifications as per Giraud et al 2023			

Management of adverse events:

Table 2: Dose Modification of avelumab for Adverse Events

Adverse reactions	**Severity***	**Recommended dose modification**
Infusion-related reactions	Grade 1	Reduce infusion rate by 50%
	Grade 2	Withhold until adverse reactions recover to Grade 0-1; restart infusion with a 50% slower rate
	Grade 3 or Grade 4	Permanently discontinue
Pneumonitis Patients should be monitored for signs and symptoms of immune-related pneumonitis	Grade 2 Suspected pneumonitis should be confirmed with radiographic imaging.	Withhold until adverse reactions recover to Grade 0-1. Corticosteroids[a] should be administered for Grade ≥ 2 events.

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and causes other than immune-related pneumonitis should be ruled out.	Grade 3 or Grade 4 or recurrent Grade 2	Permanently discontinue
Hepatitis Patients should be monitored for changes in liver function and symptoms of immune-related hepatitis and causes other than immune-related hepatitis should be ruled out.	Grade 2 or Aspartate aminotransferase (AST) or alanine aminotransferase (ALT) greater than 3 and up to 5 times upper limit of normal (ULN) or total bilirubin greater than 1.5 and up to 3 times ULN	Withhold until adverse reactions recover to Grade 0-1. Corticosteroids[a] should be administered for Grade ≥ 2 events.
	≥ Grade 3 AST or ALT greater than 5 times ULN or total bilirubin greater than 3 times ULN	Permanently discontinue
Colitis Patients should be monitored for signs and symptoms of immune-related colitis and causes other than immune-related colitis should be ruled out.	Grade 2 or Grade 3 colitis or diarrhoea	Withhold until adverse reactions recover to Grade 0-1. Corticosteroids[a] should be administered for Grade ≥ 2 events.
	Grade 4 colitis or diarrhoea or recurrent Grade 3 colitis	Permanently discontinue
Pancreatitis Patients should be monitored for signs and symptoms of immune-related pancreatitis. In symptomatic patients, obtain gastroenterology consultation and laboratory investigations (including imaging) to ensure the initiation of appropriate measures at an early stage.	Suspected pancreatitis	Withhold until resolution. Corticosteroids should be administered for immune related pancreatitis.
	Confirmed pancreatitis	Permanently discontinue
Myocarditis Patients should be monitored for signs and symptoms of immune-related myocarditis. In symptomatic patients, obtain cardiologic consultation and laboratory investigations to ensure the initiation of appropriate measures at an early stage.	Suspected myocarditis	Withhold until resolution. Corticosteroids[a] should be administered for immune-related myocarditis. If no improvement within 24 hours on corticosteroids, additional immunosuppression (e.g., mycophenolate, infliximab, anti-thymocyte globulin) should be considered.
	Confirmed myocarditis	Permanently discontinue
Endocrinopathies (hypothyroidism, hyperthyroidism) Immune-related thyroid disorders have been reported in patients	Grade 3 or Grade 4	Withhold until adverse reactions recover to Grade 0-1. Hypothyroidism should be managed

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receiving avelumab. Patients should be monitored for clinical signs and symptoms of endocrinopathies.		with replacement therapy and hyperthyroidism with anti-thyroid medicinal product as needed.
Adrenal insufficiency Immune-related adrenal insufficiencies have been reported in patients receiving avelumab. Patients should be monitored for clinical signs and symptoms of endocrinopathies.	Grade 3 or Grade 4 symptomatic	Withhold. Corticosteroids[a] should be administered for Grade ≥ 3 adrenal insufficiency followed by a taper until a dose of less than or equal to 10 mg/day has been reached.
Type 1 diabetes mellitus		Avelumab can cause Type 1 diabetes mellitus, including diabetic ketoacidosis. Patients should be monitored for hyperglycaemia or other signs and symptoms of diabetes. Initiate treatment with insulin for Type 1 diabetes mellitus
Grade ≥ 3 hyperglycaemia	Withhold	Antihyperglycaemics should be administered. Treatment with avelumab should be resumed when metabolic control is achieved on insulin replacement therapy
Nephritis and renal dysfunction	Grade 2 or 3 Serum creatinine > 1.5 - 6 x ULN	Withhold until adverse reactions recover to Grade 0-1. Corticosteroids should be administered for Grade ≥ 2 nephritis[a].
	Grade 4 Serum creatinine > 6 x ULN	Permanently discontinue
Skin reactions	Grade 3 rash	Withhold until adverse reactions recover to Grade 0-1
	Grade 4 or recurrent Grade 3 rash or confirmed Stevens–Johnson syndrome (SJS) or Toxic epidermal necrolysis (TEN)	Permanently discontinue
Other immune-related adverse reactions (including myositis, hypopituitarism, uveitis, myasthenia gravis, myasthenic syndrome, Guillain-Barré syndrome)	For any of the following: • Grade 2 or Grade 3 clinical signs or symptoms of an immune-related adverse reaction not described above	Withhold until adverse reactions recover to Grade 0-1
	For any of the following: • Life threatening or Grade 4 adverse reaction (excluding endocrinopathies controlled with hormone replacement therapy) • Recurrent Grade 3 immune-related adverse reaction	Permanently discontinue

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	• Requirement for 10 mg per day or greater prednisone or equivalent for more than 12 weeks • Persistent Grade 2 or Grade 3 immune-mediate adverse reactions lasting 12 weeks or longer	
[a]Corticosteroids should be administered for Grade ≥ 2 events (initial dose of 1 to 2 mg/kg/day prednisoLONE or equivalent followed by a corticosteroid taper). Based on the severity of the adverse reaction, avelumab should be withheld and corticosteroids administered. If corticosteroids are used to treat an adverse reaction, a taper of at least 1 month duration should be initiated upon improvement. In patients, whose immune-related adverse reactions cannot be controlled with corticosteroid use, administration of other systemic immunosuppressants may be considered.		

* Toxicity was graded per National Cancer Institute Common Terminology Criteria for Adverse Events Version 4.0 (NCI-CTCAE v4.03)

- **Immune-related adverse reactions:** Most immune-related adverse reactions with avelumab were reversible and managed with temporary or permanent discontinuation of avelumab, administration of corticosteroids and/or supportive care.

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting- Available on the NCCP website

Avelumab: Minimal **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS:

Patients have to be premedicated with an antihistamine and with paracetamol prior to the first 4 infusions of avelumab. If the fourth infusion is completed without an infusion-related reaction, premedication for subsequent doses should be administered at the discretion of the clinician.

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Table 3: Suggested pre-medications prior to avelumab infusion:

Drugs	Dose	Route
Paracetamol	1g	PO at least 30 minutes prior to avelumab infusion
Chlorphenamine	10mg	IV bolus at least 30 minutes prior to avelumab infusion

OTHER SUPPORTIVE CARE:

Women of childbearing potential should be advised to avoid becoming pregnant while receiving avelumab and should use effective contraception during treatment with avelumab and for at least 1 month after the last dose of avelumab.

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics for full details of adverse effects.
- Please note the NCCP National SACT regimens do not provide information regarding black triangle (▼) status. Please refer to the relevant Summary of Product Characteristics (SmPC) for this information.

REGIMEN SPECIFIC COMPLICATIONS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

SUPPORT RESOURCES/Useful Links:

Please note that this is for information only and does not constitute endorsement by the NCCP

Patient alert card:

https://assets.hpra.ie/products/Human/10498/6b050953-0881-41e7-abbb-7475cb4c4622.pdf

REFERENCES:

1. Kaufman et al. Updated efficacy of avelumab in patients with previously treated metastatic Merkel cell carcinoma after ≥1 year of follow-up: JAVELIN Merkel 200, a phase 2 clinical trial. Journal for ImmunoTherapy of Cancer (2018) 6:7
2. Kaufman et al. Avelumab in patients with chemotherapy-refractory metastatic Merkel cell carcinoma: a multicentre,single-group, open-label, phase 2 trial Lancet Oncol 2016; 17: 1374–85

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4. Powles T, Park SE, Voog E, et al. Avelumab first-line (1L) maintenance for advanced urothelial carcinoma (UC): Long-term follow-up results from the JAVELIN Bladder 100 trial. Presented at: ASCO-GU 2022; February 17-19, 2022; Abstract 487
5. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
6. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://healthservice.hse.ie/documents/7152/Classification_Document_for_Systemic_Anti-_Cancer_Therapy_SACT_Induced_Nausea__pWgDCrm.pdf
7. Avelumab (Bavencio®) 20 mg/mL concentrate for solution for infusion Summary of Product characteristics. Last updated 22/01/2026. Accessed March 2026. Available at: https://www.ema.europa.eu/en/documents/product-information/bavencio-epar-product-information_en.pdf

From 20/05/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie

Version	Date	Amendment	Approved By
1	10/04/2019		Dr Deirdre O'Mahony
2	09/07/2019	Updated immune related adverse reactions to include pancreatitis as per SPC update	Dr Deirdre O'Mahony
3	14/02/2020	Updated dosing posology as per SmPC update to flat dosing. Updated dose modification and adverse events for pancreatitis and myocarditis as per SmPC update	Dr Deirdre O'Mahony
4	20/10/2020	Added support resources	Prof Maccon Keane
5	28/04/2021	Reviewed. Updated Table 2 (Dose modification for adverse events) as per SPC update. Updated Table 5 (management of immune related adverse effects) as per SPC update. Updated support resources.	Prof Maccon Keane
6	01/09/2022	Addition of new indication.	Prof Maccon Keane
6a	08/05/2025	Update to ICD-10 code.	NCCP
6b	14/08/2025	Updated exclusions section with information regarding previous PDL1	NCCP

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		treatment. Added links to emetogenic potential section.	
7	29/06/2026	Regimen reviewed. Added cautions section. Updated exclusions section. Updated renal and and hepatic dose modifications table to align with Giraud et al 2023.Updated in line with NCCP standardisation. Updated table 2 to include all adverse events information. Update to review schedule.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie

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